

Gate -2011-IN -16 Question

QUESTION

For the Boolean expression $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$, the minimized Product of Sum (PoS) expression is

(A) $f = (b + \bar{c}).(a + \bar{c})$

(B) $f = (\bar{b} + c).(\bar{a} + c)$

(C) $f = (\bar{b} + c).(a + \bar{c})$

(D) $f = \bar{c} + abc$

QUESTION ANALYSIS

Solution

Assuming the Boolean expression is

$$f = \bar{a}b\bar{c} + a\bar{b}\bar{c} + \bar{a}bc + abc + ab\bar{c}$$

Step 1: Convert to Minterms

$$\bar{a}b\bar{c} = m_2$$

$$\bar{a}bc = m_3$$

$$a\bar{b}\bar{c} = m_4$$

$$ab\bar{c} = m_6$$

$$abc = m_7$$

Therefore,

$$f = \Sigma m(2, 3, 4, 6, 7)$$

Step 2: Find the Zero Cells

The missing minterms are

$$m_0, m_1, m_5$$

Hence,

$$f = \Pi M(0, 1, 5)$$

Step 3: K-map Grouping of Zeros

AB\C	0	1
00	0	0
01	1	1
11	1	1
10	1	0

Group 1: m_0 and m_1 Common variables:

$$a = 0, \quad b = 0$$

Maxterm:

$$(a + b)$$

Group 2: m_1 and m_5 Common variables:

$$b = 0, \quad c = 1$$

Maxterm:

$$(b + \bar{c})$$

Step 4: Write the Minimized PoS Expression

$$f = (a + b)(b + \bar{c})$$

Verification

$$(a + b)(b + \bar{c})$$

is equal to 0 for

$$000, 001, 101$$

and equal to 1 for

$$010, 011, 100, 110, 111$$

which exactly matches the given function.

$$\text{Minimized PoS} = (a + b)(b + \bar{c})$$

Components Required

S.No	Component	Quantity
1	Pygmy FPGA Board	1
2	7447 BCD-to-Seven-Segment Decoder IC	1
3	Seven-Segment Display	1
4	USB Cable	1
5	Computer/Laptop (Linux/Ubuntu/Termux)	1
6	FPGA Toolchain (QL SymbiFlow)	1
7	Breadboard	1
8	Connecting Wires	As Required

Hardware Connections

Pygmy FPGA Pin	7447 IC Pin	Function
15	A	D0 (LSB) Input to 7447
16	B	D1 Input to 7447
17	C	D2 Input to 7447
18	D	D3 (MSB) Input to 7447
11	Input A	Boolean Function Input A
12	Input B	Boolean Function Input B
13	Input C	Boolean Function Input C

7447 IC Pin	Seven-Segment Display	Function
a	Segment a	Top Segment
b	Segment b	Upper Right Segment
c	Segment c	Lower Right Segment
d	Segment d	Bottom Segment
e	Segment e	Lower Left Segment
f	Segment f	Upper Left Segment
g	Segment g	Middle Segment
COM	+5V	Common Anode Connection

Power Connections	Connection
7447 VCC	+5V Supply
7447 GND	Ground
Seven-Segment COM	+5V
Pygmy GND	Common Ground with 7447 Circuit

Note:

- Pygmy FPGA pins 11, 12, and 13 are used as inputs corresponding to variables A , B , and C .
- The FPGA evaluates the Boolean function and generates a 4-bit BCD output on pins 15, 16, 17, and 18 (D_0 , D_1 , D_2 , and D_3).

- The 7447 IC decodes the BCD input and drives the seven-segment display.
- Since the 7447 is designed for common-anode displays, the COM pin of the display is connected to $+5V$.
- All grounds must be connected together to establish a common reference.

Execution Procedure

1. Create the Verilog source file *lcd_task.v* and the constraint file

1. Compile the design using the QL SymbiFlow toolchain

```
ql_symbiflow -compile \
-d ql-eos-s3 \
-P pu64 \
-v lcd_task.v \
-t top \
-p lcd_task.pcf
```

2. After successful compilation, the bitstream file is generated inside the build directory.

```
build/top.bit
```

3. Rename the generated .bit file to a .bin file:

```
mv build/top.bit top.bin
```

4. Copy the generated .bin file to the computer where the FPGA programming environment has been established.

5. Open a terminal and navigate to the directory containing the generated binary file:

```
cd <path_to_binary_file>
```

6. Put the Pygmy FPGA board into Flash Mode and connect it to the computer using a USB cable.

7. Verify that the board is detected by the system:

```
ls -l /dev/ttyACM0
```

If the device is detected, information about the serial port will be displayed.

8. Program the FPGA board using the TinyFPGA programmer utility:

```
python3 tinyfpga-programmer-gui.py \  
--port /dev/ttyACM0 \  
--m4app boolean_display_0.bin \  
--node n4
```

9. Wait until the programming process completes successfully.
10. Press the RESET button on the Pygmy FPGA board.
11. Observe the output on the hardware and verify the expected functionality.

Results and Observations

The designed Boolean function was successfully implemented on the Pygmy FPGA board and tested using a seven-segment display interfaced through the 7447 BCD-to-Seven-Segment Decoder IC. The FPGA received the input combinations through pins 11, 12, and 13 and generated the corresponding BCD output on pins 15, 16, 17, and 18. The 7447 decoder converted the BCD output into appropriate segment signals to display the digits 0 and 1.

Figure ?? shows the hardware setup and the output displayed on the seven-segment display during experimental verification.

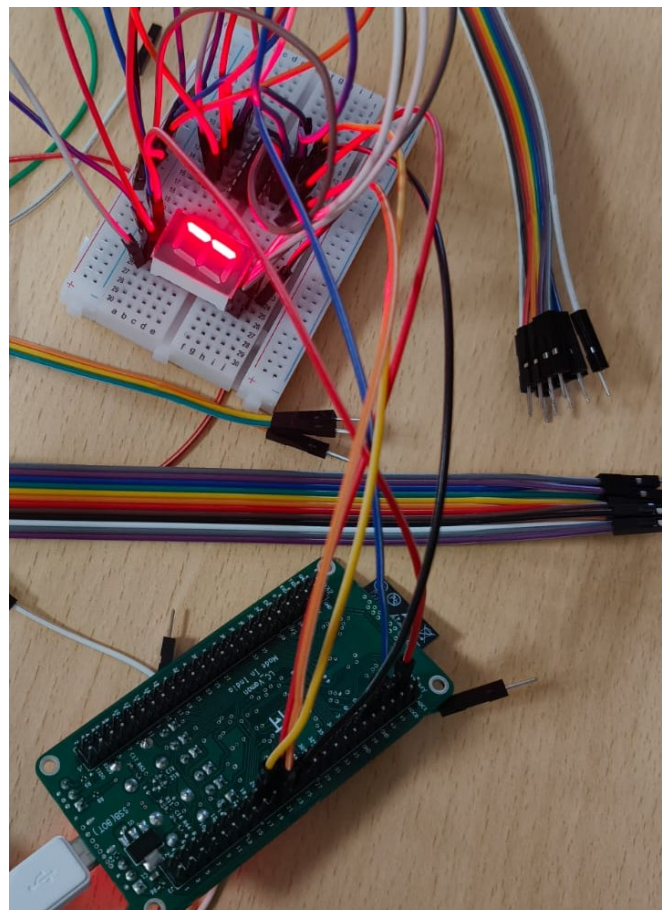


Figure 1: Output Displayed on the Seven-Segment Display Using 7447 Decoder

The output obtained from the hardware implementation was verified against the theoretical truth table. For each combination of inputs A , B , and C , the displayed digit corresponded to the expected output of the Boolean function.

Observed Truth Table

A	B	C	Displayed Digit
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1